evaluates threats, selects optimal flight paths, and manages weapons deployment under human supervision.

3. *Integrated sensor and communication systems*: Equipped with advanced sensors for target detection, terrain following, and situational awareness, Taranis maintains real-time communication links with ground stations for mission updates and data exchange.
4. *Internal weapon bay*: To maintain stealth, weapons such as precision-guided munitions (PGMs) are carried internally. While Taranis is a demonstrator, it is designed to accommodate 250–500 kg class smart bombs for strike missions.
5. *Collaborative engineering effort*: The program involved major UK defense partners, including Rolls-Royce (engine systems), QinetiQ (flight testing and sensors), and GE Aviation (avionics), establishing a comprehensive domestic UAV capability.

Key specifications:

- *Type*: Unmanned combat aerial vehicle (UCAV) – technology demonstrator
- *Developer*: BAE Systems, United Kingdom
- *First flight*: August 2013
- *Wingspan*: 9.9 m (32 ft)
- *Length*: 11.3 m (37 ft)

- *Propulsion*: Rolls-Royce Adour Mk 951 turbofan engine
- *Speed*: High subsonic (~1000 km/h)
- *Ceiling*: 40,000 ft (12,200 m)
- *Range*: Classified (~3000 km estimated)
- *Payload*: Internal bay for 2 × 250–500 kg-class smart bombs